

AKSHANSH PAREEK

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Delhi, India

RESEARCH FOCUS

I work on Human-AI Interaction, Human-Agent Interaction, and community participatory systems. My work combines interaction design, UX research, computational social science, and data science methods to study how AI systems communicate values, shape decisions, and can be made more transparent, context-aware, and socially grounded, and aligned with human intent in creative and enterprise settings.

EDUCATION

Delhi Technological University (DTU)

Bachelor of Design

Delhi, India

First Class with Distinction; CGPA: 8.38/10.0

Thesis: Measuring Regional and Occupational Bias in LLM's. Grade: A+

RESEARCH EXPERIENCE

Sentient Futures

Remote

Research Fellow; Supervisor: Julia Bossman

February 2026 – Now

- Engineered a controlled LoRA fine-tuning experiment on the Meta-Llama-3-8B base model to quantify the AI alignment tax, demonstrating that warmth-tuned models exhibited up to a 46.1 percentage point increase in strict factual abandonment under confident user pressure.
- Curated a 2,000-example balanced training dataset isolating emotional validation from factual content, utilizing VADER sentiment analysis and a Qwen3-32B-Instruct LLM judge for strict data quality control.
- Developed a 120-item multi-domain evaluation suite to assess model robustness across soft, confident, and emotional social pressure conditions, identifying a novel 64.3% "affective deflection" failure mode.
- Formulated a custom multi-class labeling scheme and conducted paired item-level statistical analysis, proving the treatment model retained a high 81.7% baseline factual accuracy despite explicitly adopting user-suggested incorrect answers under pressure

University of Cambridge

Remote

Research Assistant; Advisor: Prof. Alan Blackwell

August 2025 – December 2025

- Directed the research design for an investigation into institutional and geographic bias in academic publishing, developing core hypotheses, experimental protocols, and ethics-aligned deliverables to ensure methodological rigor.
- Engineered controlled peer-review experiments by systematically manipulating author affiliations, establishing a framework to isolate and measure systemic disparities in manuscript submission treatment.
- Analyzed experimental data utilizing logistic regression to estimate evaluation gaps, successfully quantifying acceptance rate disparities across varied institutional and geographic conditions.

MIT Media Lab

Remote

Research Assistant; Advisor: Hope Schroeder, Dr. Solon Barocas

October 2024 – May 2025

- Conducted a mixed-methods empirical review of 82 positionality statements extracted from 7 years (2018–2024) of ACM FAccT proceedings to evaluate the efficacy of policy interventions on responsible computing practices.
- Formulated 3 comprehensive qualitative annotation schemes comprising 26 distinct labels, achieving high inter-rater reliability (Krippendorff's $\alpha = 0.91$) to systematically categorize researcher disclosures across identity, geography, and institutional contexts.
- Modeled the causal impact of conference policy changes using OLS regression and two-sample z-tests, demonstrating a statistically significant increase in positionality statement adoption among non-community-focused papers ($p < 0.01$).
- Synthesized findings to expose field-wide trends of "disclosure without engagement," proposing 3 data-driven policy recommendations to restructure peer-review anonymity guidelines.

University of Bristol

Research Assistant; Advisor: Dr. Richard Cole

Remote

August 2024 – September 2024

- Investigated the technical limitations of LLM-powered game agents, evaluating literature on NPC behavior, memory, and hallucination risks to establish baseline parameters for the research.
- Constructed a quantitative risk matrix to systematically evaluate algorithmic coherence failures, setting clear psychological impact benchmarks for the study.
- Delivered an R&D report and presented actionable recommendations to external project partners regarding the integration and risks of generative AI in gaming environments.

University of Manitoba, Dept. of Computer Science

Research Assistant; Advisor: Prof. Celine Latulipe

Remote

September 2023 – June 2024

- Designed custom 3D assets, scenes, and an interactive virtual gallery to host interior design exhibitions, successfully facilitating a simulated gallery visit and online public opening event.
- Assessed platform engagement, accessibility, and user social interaction by systematically extracting and processing feedback from visitor surveys, participant interviews, and expert curator reviews.
- Co-authored a peer-reviewed research paper evaluating the efficacy of web-based 3D exhibition environments in design education, resulting in an official acceptance to the Graphics Interface conference.

Concordia University

HCI Research Intern

Remote

September 2023 – January 2024

- Evaluated existing literature on generative AI within interaction design—specifically analyzing LLM-assisted workflows and visual generation tools, establishing foundational methodologies for the research team.
- Optimized the Edgy AI Chatbot's conversation flow, tone, and error-recovery protocols by orchestrating targeted usability tests and integrating quantitative task analytics with qualitative user feedback.

Indian Institute of Technology Guwahati (HCI LAB)

UX Research Intern; Advisor: Prof. Debayan Dhar and Dr. Neelarnab Dutta

Guwahati, India

May 2023 – July 2023

- Examined product design frameworks and prevailing market trends within the medical sector to identify viable pathways for medical device development and commercialization.
- Authored an instructional workshop manual detailing design and marketing best practices, equipping emerging entrepreneurs with actionable strategies for launching new medical products..

WORK EXPERIENCE

Divergent Classes

Delhi, India

- *Data Consultant*

April 2025 – February 2026

- Programmed an automated learning analytics pipeline using Python and pandas, aggregating granular metrics—including mock test scores, topic-wise accuracy, and time-per-question—to track the academic trajectories of 200+ students.
- Devised predictive risk-scoring models to identify early signs of student underperformance, directly enabling instructors to execute targeted interventions and maintain consistent support across diverse cohorts.

RESEARCH PUBLICATIONS

- Schroeder, H., **Pareek, A.**, & Barocas, S., Disclosure without Engagement: An Empirical Review of Positionality Statements at FAccT." *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency (FAccT '25)*, <https://doi.org/10.1145/3715275.3732079>.
- **Pareek, A.**, Goswami, T., & Singh, R., "Creativity - ERROR 404: Navigating Creativity and Ownership in the Age of AI-Assisted Students" Manuscript accepted at *International Conference of engineering design 2025*. [[Preprint](#)]
- **Co-author.** "Exploring the Use of a Web-Based 3D Virtual Environment for Interior Design Work Exhibitions" accepted at *Graphics Interface Conference* [[Preprint](#)].

PROJECTS

Creativity - ERROR 404: Human-AI Collaboration Study

Project Lead/HCI Researcher; Collaborator: Tanishq Goswami; Advisor: Prof. Ravindra Singh

- Executed a multi-stage qualitative study involving semi-structured interviews with 20 undergraduate design and engineering students to evaluate the impact of generative AI on creative independence and authorship.
- Systematized transcript data utilizing collaborative reflexive thematic analysis in Figma Jam, establishing an activity-based task taxonomy across 6 creative domains and validating findings through rigorous analyst triangulation and saturation memos.
- Introduced a 5-pillar 'Compact Framework of Ethical Integration Principles' alongside an actionable ownership responsibility matrix to strictly guide disclosure, boundaries, and reflective documentation in human-AI collaboration.

FinEquity: Participatory Algorithm Auditing for Microfinance

Project Lead/HCI Researcher; Collaborator: Ragini Mishra; Advisor: Prof. Ravindra Singh

- Initiated the deployment of FinEquity, a participatory algorithm auditing platform enabling Self-Help Group (SHG) members and marginalized handloom workers to uncover and document systemic microfinance lending bias.
- Administered a comprehensive mixed-methods UX research protocol—integrating in-depth interviews, diary studies, contextual observations, and user surveys—to refine user flows and validate platform accessibility, achieving a high 80.6/100 System Usability Scale (SUS) score.
- Quantified hard-coded occupational discrimination against handloom workers utilizing Chi-Square testing ($p = 0.002$), successfully proving a 20-point systemic approval barrier regardless of user creditworthiness.

VoiceWeave: Detecting Contribution Suppression

Machine Learning Researcher; Advisor: Prof. Partha Pratim Das

- Compiled a novel, large-scale corpus comprising 38,000 annotated dialogue turns from 850+ multi-party conversations, establishing a highly reliable ($k=0.81$) data baseline for quantifying participation inequality.
- Trained an end-to-end Random Forest classifier utilizing a 4-metric interpretable feature pipeline—intentionally bypassing opaque semantic embeddings—to predict voice suppression with 85% accuracy, 87% precision, and a 92% ROC-AUC.
- Translated predictive outputs into an interactive web application utilizing SHAP partial dependence analysis, mechanistically proving how participation imbalance drives suppression risk and equipping facilitators with real-time intervention insights.

SERVICE

Volunteer Mentor | *Desh Ke Mentor Programme, Government of Delhi*

Delhi, India

Mentor to an assigned high school student

May 2022 – January 2023

- Counseled student through consistent bi-weekly mentoring sessions, delivering individualized career advice and long-term academic trajectory planning.
- Tutored mentee in core high school subjects to bridge foundational knowledge gaps.

Public Relations Lead *Entrepreneurship Cell (E-Cell)*

Delhi, India

October 2023 – May 2025

- Orchestrated a 300+ attendee entrepreneurial panel discussion and managed guest outreach, managed 12+ high-profile startup founders and venture capitalists during flagship campus events. Expanded the inter-college entrepreneurial network by 25%.

SKILLS

Programming & Software: Python, Node.js, pandas, NumPy, scikit-learn, TensorFlow, Git/GitHub, Figma.

AI & Systems: OpenAI API, prompt engineering, Model fine tuning.

Qualitative & HCI Methods: Semi-structured interviews, thematic analysis, qualitative coding, rapid prototyping, UX auditing, Ethnographic studies.

Certifications: Human-Computer Interaction & UX Design (IIT Guwahati) • AI Governance (Blue Dot Impact).

Languages: English, Hindi (native), Gujarati.